**University of Khartoum**
Faculty of Mathematical Sciences and Informatics

**GridGuard Khartoum**

# AI-Based Smart Load Shedding and Energy Management System for Developing Countries

Graduation project - Bachelor of Computer Science

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**Application interface - 21 captured screens**

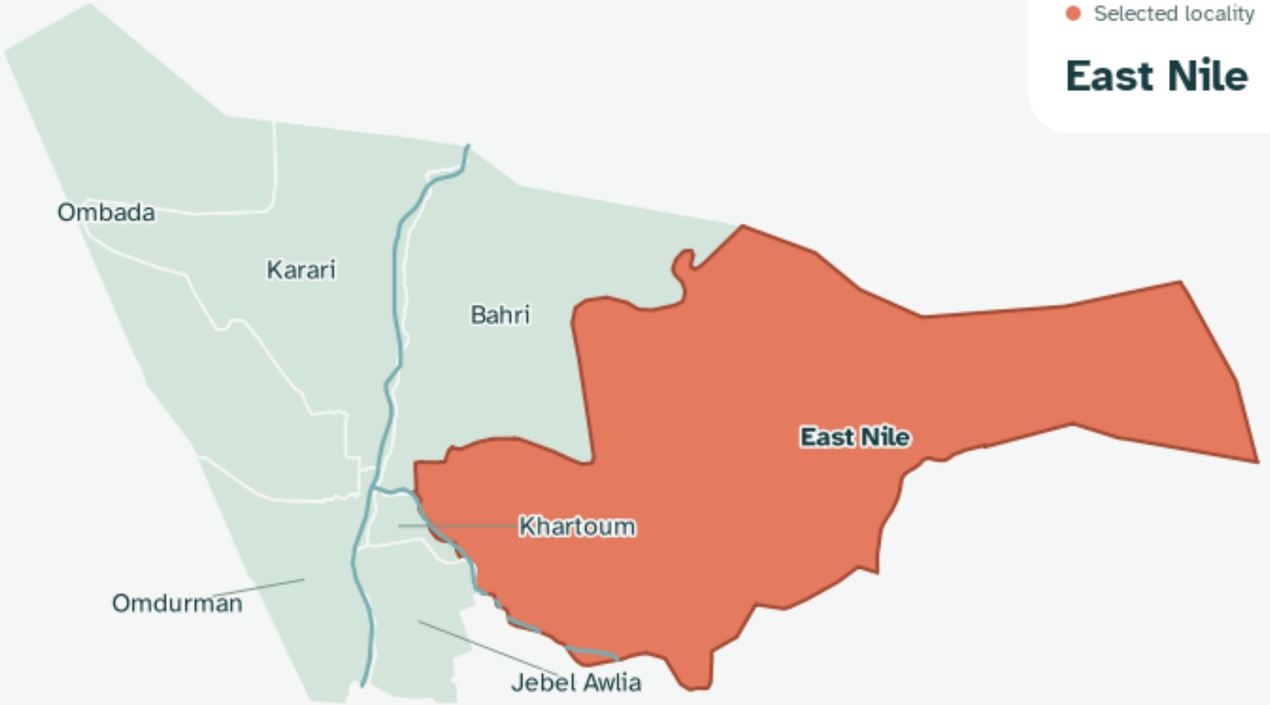
# Khartoum.

## Power allocation.
## Seven localities in view.

Explore the current trained PPO, Fixed Rotation and constrained optimizer on one live scenario, or compare their frozen study results.

**Explore scenario** **Compare results**

Choose a shortage level and seed to begin.




Selected locality

**East Nile**

| 7 | 24 | 5 |
|---|---|---|
| Localities | Hours | Service sectors |

**↗ PPO**

Learned controller

**↻ Fixed Rotation**

Baseline controller

One scenario.
A matched comparison.

Synthetic scenarios · Decision support only

Boundaries: OCHA COD-AB 2020; Rivers: Natural Earth 5.0.
Geographic context, not an electrical network.

# Khartoum, hour by hour.

No live scenario has been run. Choose inputs, then run a scenario.

Current trained PPO · ready to run
Synthetic scenario · Decision support only

Shortage level: Moderate

Seed: 50002

PPO | Fixed Rotation | Optimization

Run scenario

Full Service | Critical Only | Off | Selected locality




Boundaries: OCHA COD-AB 2020; Rivers: Natural Earth 5.0.
Geographic context, not an electrical network.

\+ | − | Fit

Selected locality —

## East Nile

Not run

| Power served | Demand |
|---|---|
| — | — |

| Sector service | Demand met |
|---|---|
| Hospital | — |
| Water | — |
| Bakery | — |
| Residential | — |
| Commercial | — |

Next hour: unavailable · display only

▶ —

No scenario yet

00:00 | 04:00 | 08:00 | 12:00 | 16:00 | 20:00 | 23:00

Available —

Total demand —

## The full 24 hours

PPO above; Fixed Rotation below.
The same scenario.

| | Served energy | Critical service | Jain fairness | Supply utilization | Daily reward |
|---|---|---|---|---|---|
| | — | — | — | — | — |
| Fixed Rotation | — | — | — | — | — |

# Khartoum, hour by hour.

Explore how power reaches each locality.

PPO · seed 8111 · iteration 2500
Synthetic scenario · Decision support only

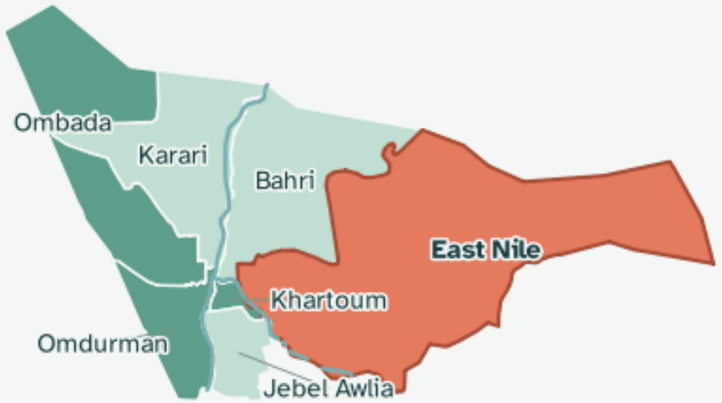
Shortage level: Moderate

Seed: 501

PPO | Fixed Rotation | Optimization

Run scenario

Full Service | Critical Only | Off | Selected locality




Boundaries: OCHA COD-AB 2020; Rivers: Natural Earth 5.0.
Geographic context, not an electrical network.

+ | − | Fit

Selected locality 12:00

## East Nile

Critical Only

| Power served | Demand |
|---|---|
| 13.3 MW | 56.6 MW |

| Sector service | Demand met |
|---|---|
| Hospital | 100.0% |
| Water | 100.0% |
| Bakery | 100.0% |
| Residential | 0.0% |
| Commercial | 0.0% |

Next-hour estimate: 56.1 MW
Display only

▶ 12:00
Hour 12 of 24

00:00 04:00 08:00 12:00 16:00 20:00 23:00

Available 271.7 MW
Total demand 429.4 MW

## The full 24 hours

PPO above;
Optimization below. Same scenario.

| Served energy | Critical service | Jain fairness | Supply utilization | Daily reward |
|---|---|---|---|---|
| 6,575 MWh | 100.0% | 97.7% | 97.7% | 23.5442 |
| Optimization 6,604 MWh | Optimization 100.0% | Optimization 95.0% | Optimization 98.2% | Optimization 23.4625 |

# Khartoum, hour by hour.

Explore how power reaches each locality.

PPO · seed 8111 · iteration 2500
Synthetic scenario · Decision support only

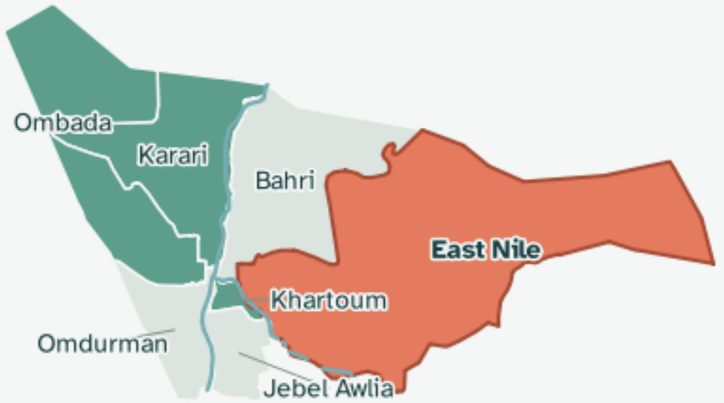
Shortage level: Moderate

Seed: 501

PPO | Fixed Rotation | Optimization

Run scenario

Full Service | Critical Only | Off | Selected locality




Boundaries: OCHA COD-AB 2020; Rivers: Natural Earth 5.0.
Geographic context, not an electrical network.

+ | − | Fit

Selected locality 12:00

## East Nile

Off

| Power served | Demand |
|---|---|
| 0.0 MW | 56.6 MW |

| Sector service | Demand met |
|---|---|
| Hospital | 0.0% |
| Water | 0.0% |
| Bakery | 0.0% |
| Residential | 0.0% |
| Commercial | 0.0% |

Next-hour estimate: 56.1 MW
Display only

▶ 12:00
Hour 12 of 24

00:00 04:00 08:00 12:00 16:00 20:00 23:00

Available 271.7 MW
Total demand 429.4 MW

## The full 24 hours

Fixed Rotation above;
PPO below. Same scenario.

| Served energy | Critical service | Jain fairness | Supply utilization | Daily reward |
|---|---|---|---|---|
| 4,640 MWh | 43.4% | 87.8% | 68.9% | 13.9951 |
| PPO 6,575 MWh | PPO 100.0% | PPO 97.7% | PPO 97.7% | PPO 23.5442 |

# Khartoum, hour by hour.

Explore how power reaches each locality.

PPO · seed 8111 · iteration 2500
Synthetic scenario · Decision support only

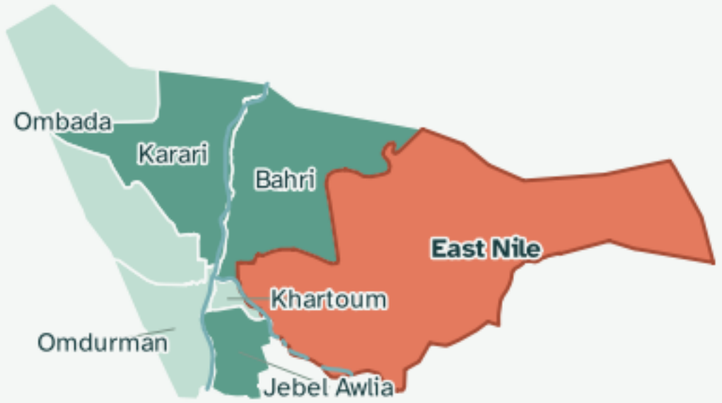
Shortage level: Moderate

Seed: 501

PPO | Fixed Rotation | Optimization

Run scenario

Full Service | Critical Only | Off | Selected locality




Boundaries: OCHA COD-AB 2020; Rivers: Natural Earth 5.0.
Geographic context, not an electrical network.

+ − Fit

Selected locality 12:00

## East Nile

Full Service

| Power served | Demand |
|---|---|
| 56.6 MW | 56.6 MW |

| Sector service | Demand met |
|---|---|
| Hospital | 100.0% |
| Water | 100.0% |
| Bakery | 100.0% |
| Residential | 100.0% |
| Commercial | 100.0% |

Next-hour estimate: 56.1 MW
Display only

▶ 12:00
Hour 12 of 24

00:00 04:00 08:00 12:00 16:00 20:00 23:00

Available 271.7 MW
Total demand 429.4 MW

## The full 24 hours

Optimization above;
PPO below. Same scenario.

| Served energy | Critical service | Jain fairness | Supply utilization | Daily reward |
|---|---|---|---|---|
| 6,604 MWh | 100.0% | 95.0% | 98.2% | 23.4625 |
| PPO 6,575 MWh | PPO 100.0% | PPO 97.7% | PPO 97.7% | PPO 23.5442 |

# Results

Earlier results | Study details

## Current study

Current study | Follow-up comparison

960 matched conditions · 320 base draws · five independent PPO runs

Critical interruption

**0.00 locality-h**

PPO · five-run mean

Fixed Rotation 102.05 locality-h · Optimizer 0.00 locality-h

Jain service fairness

**97.02%**

PPO · five-run mean

Fixed Rotation 87.76% · Optimizer 94.78%

Supply utilization

**97.02%**

PPO · five-run mean

Fixed Rotation 71.23% · Optimizer 97.41%

| Outcome | Fixed Rotation | Constrained optimizer | PPO |
|---|---|---|---|
| Critical interruption | 102.05 locality-h | 0.00 locality-h | 0.00 locality-h |
| Jain service fairness | 87.76% | 94.78% | 97.02% |
| Supply utilization | 71.23% | 97.41% | 97.02% |
| Daily reward | 13.9688 | 23.4468 | 23.4967 |
| Switching rate | 25.00% | 20.55% | 27.76% |

**Against the constrained optimizer: fairness +2.237 pp · utilization −0.399 pp · daily reward +0.0499.**

# Results

## Follow-up comparison

Current study | Follow-up comparison

**Same 960 matched conditions · 320 base draws · five frozen selected PPO policies**

Post-hoc follow-up · previously evaluated scenarios · no retraining or new confirmatory claim.

Controller means | Paired differences | Critical shortfall

Critical-aware Fixed Rotation reserves critical demand, then rotates discretionary priority without reward optimization. Fairness Rule uses the frozen imitation-teacher setting. These comparisons do not isolate learning alone.

| Controller | Served MWh | Critical locality-h | Fairness % | Utilization % | Daily reward | Switching % |
|---|---|---|---|---|---|---|
| Fixed Rotation | 4551.33 | 102.05 | 87.76 | 71.23 | 13.9688 | 25.00 |
| Critical-aware Fixed Rotation | 6286.39 | 0.00 | 94.95 | 93.38 | 23.2257 | 22.77 |
| Constrained Optimization | 6547.90 | 0.00 | 94.78 | 97.41 | 23.4468 | 20.55 |
| Fairness Rule | 6544.70 | 0.00 | 96.15 | 97.32 | 23.4865 | 24.84 |
| PPO | 6528.02 | 0.00 | 97.02 | 97.02 | 23.4967 | 27.76 |

**PPO versus Fairness Rule is a tradeoff: daily reward +0.0102, served energy −16.68 MWh/day. Reward is dimensionless; it cannot be converted into energy or cost.**

# Results

## Follow-up comparison

Current study | Follow-up comparison

**Same 960 matched conditions · 320 base draws · five frozen selected PPO policies**

Post-hoc follow-up · previously evaluated scenarios · no retraining or new confirmatory claim.

Controller means | Paired differences | Critical shortfall

Descriptive pointwise paired 95% intervals; pp = percentage points. All five PPO runs receive equal weight. Original primary criteria remain unchanged.

### PPO vs Critical-aware Fixed Rotation

| Outcome | PPO difference | 95% interval |
|---|---|---|
| Served energy | +241.63 | [+236.89, +245.91] MWh |
| Fairness | +2.07 | [+1.96, +2.17] pp |
| Utilization | +3.64 | [+3.57, +3.70] pp |
| Daily reward | +0.2710 | [+0.2631, +0.2782] |

### PPO vs Fairness Rule

| Outcome | PPO difference | 95% interval |
|---|---|---|
| Served energy | −16.68 | [−20.84, −12.99] MWh |
| Fairness | +0.86 | [+0.75, +0.98] pp |
| Utilization | −0.31 | [−0.36, −0.26] pp |
| Daily reward | +0.0102 | [+0.0024, +0.0172] |

# Results

## Follow-up comparison

Current study | Follow-up comparison

**Same 960 matched conditions · 320 base draws · five frozen selected PPO policies**

Post-hoc follow-up · previously evaluated scenarios · no retraining or new confirmatory claim.

Controller means | Paired differences | Critical shortfall

Supply is 75% or 25% of critical demand: the critical floor is physically infeasible. Critical-only fallback shares the shortage; it is not uninterrupted critical service. Each stress family has 320 conditions and stays separate from nominal results.

| Supply / critical | Controller | Fallback hours | Interrupted locality-h | Zero critical locality-h | Unmet critical MWh | Power violation rate |
|---|---|---|---|---|---|---|
| 75% | Fixed Rotation | 0 | 168 | 96 | 2163.33 | 0 |
| 75% | Critical-aware Fixed Rotation | 24 | 168 | 0 | 671.51 | 0 |
| 75% | Constrained Optimization | 24 | 168 | 0 | 671.51 | 0 |
| 75% | Fairness Rule | 24 | 168 | 0 | 671.51 | 0 |
| 75% | PPO | 24 | 168 | 0 | 671.51 | 0 |
| 25% | Fixed Rotation | 0 | 168 | 96 | 2511.80 | 0 |
| 25% | Critical-aware Fixed Rotation | 24 | 168 | 0 | 2014.53 | 0 |
| 25% | Constrained Optimization | 24 | 168 | 0 | 2014.53 | 0 |
| 25% | Fairness Rule | 24 | 168 | 0 | 2014.53 | 0 |
| 25% | PPO | 24 | 168 | 0 | 2014.53 | 0 |

# Current study

960 matched conditions · 320 base draws · five independent PPO runs

PPO vs Fixed Rotation | PPO vs optimizer | Individual runs | Shortage levels | Evidence

Paired 97.5% intervals adjusted for the two primary comparisons. pp = percentage points; PPO means weight all five runs equally.

| Outcome | PPO difference | 97.5% interval |
|---|---|---|
| Critical interruption | −102.0500 | [−102.0812, −102.0250] locality-h |
| Jain service fairness | +9.2600 | [+9.1423, +9.3712] pp |
| Supply utilization | +25.7815 | [+25.7017, +25.8594] pp |
| Daily reward | +9.5279 | [+9.5184, +9.5357] |
| Switching rate | +2.7639 | [+2.0601, +3.6639] pp |

Fixed Rotation uses Off/Full Service without critical reservation. This compares complete systems; it does not isolate learning alone.

Close

# Current study

960 matched conditions · 320 base draws · five independent PPO runs

PPO vs Fixed Rotation | PPO vs optimizer | Individual runs | Shortage levels | Evidence

Paired 97.5% intervals adjusted for the two primary comparisons. pp = percentage points; PPO means weight all five runs equally.

| Outcome | PPO difference | 97.5% interval |
|---|---|---|
| Critical interruption | +0.0000 | [+0.0000, +0.0000] locality-h |
| Jain service fairness | +2.2368 | [+2.0955, +2.3731] pp |
| Supply utilization | −0.3989 | [−0.4642, −0.3408] pp |
| Daily reward | +0.0499 | [+0.0406, +0.0577] |
| Switching rate | +7.2183 | [+6.5310, +8.1124] pp |

Both protect critical demand. The optimizer maximizes the current hour; the result does not compare PPO with a full-day optimum.

Close

# Current study

960 matched conditions · 320 base draws · five independent PPO runs

PPO vs Fixed Rotation | PPO vs optimizer | Individual runs | Shortage levels | Evidence

Frozen final-test means on matched scenarios.

| PPO seed | Selected iteration | Fairness | Utilization | Reward |
|---|---|---|---|---|
| 1729 | 2900 | 96.89% | 97.08% | 23.4835 |
| 2718 | 3000 | 96.87% | 97.07% | 23.4913 |
| 3141 | 2900 | 97.12% | 97.01% | 23.5034 |
| 5772 | 3000 | 97.08% | 96.99% | 23.5014 |
| 8111 | 2500 | 97.14% | 96.92% | 23.5040 |

Close

# Current study

960 matched conditions · 320 base draws · five independent PPO runs

PPO vs Fixed Rotation | PPO vs optimizer | Individual runs | Shortage levels | Evidence

Frozen final-test means on matched scenarios.

| Shortage | Controller | Fairness | Utilization | Reward |
|---|---|---|---|---|
| Mild | Fixed Rotation | 87.74% | 54.28% | 13.1911 |
| Mild | Constrained optimizer | 96.41% | 98.09% | 23.5881 |
| Mild | PPO | 97.99% | 98.12% | 23.6098 |
| Moderate | Fixed Rotation | 87.74% | 68.69% | 13.9600 |
| Moderate | Constrained optimizer | 95.00% | 98.02% | 23.4626 |
| Moderate | PPO | 97.21% | 97.52% | 23.5002 |
| Severe | Fixed Rotation | 87.80% | 90.73% | 14.7554 |
| Severe | Constrained optimizer | 92.94% | 96.13% | 23.2897 |
| Severe | PPO | 95.86% | 95.40% | 23.3801 |

Close

# Current study

960 matched conditions · 320 base draws · five independent PPO runs

PPO vs Fixed Rotation | PPO vs optimizer | Individual runs | Shortage levels | Evidence

## Pinned evidence and independent replay

3,000 iterations per run · 92.16 million training transitions total

PPO starts from fairness-rule imitation, then trains with critical-service constraints.

Zero critical interruptions and power-balance violations for PPO and the constrained optimizer.

Study statistics were verified on CUDA. Independent replay checked every input hash, 2,304 final trajectories and 5,280 validation trajectories with zero discrepancy.

**data/evidence/focused-ppo-study.json**

SHA-256 · a4d5a849612f06eedef96e0d28e08247e5e3757eba0b4f2e4f1a893e9d4b8247

**data/evidence/focused-ppo-training.json**

SHA-256 · e4f8b8f7c73693f713ecef69a06434ce75a3a101d2541654d7009398c4eb5624

**docs/focused-ppo-validation.json**

SHA-256 · df1cc59a3a1cd94f8333f77693dc47ef1afb6007093d940dd9deab06383a1fe1

Live Scenario uses one validation-selected PPO policy from this training study with the same critical-service constraints and constrained optimizer. A single scenario can differ from these frozen five-run benchmark means.

Close

Earlier frozen policy · separate study and scenarios

# Results

Frozen holdout results

PPO and Fixed Rotation on identical matched scenarios.

Evidence details | Extended study

960 matched scenario conditions  from 320 independent base draws

Mild, Moderate and Severe shortage

## Critical-outage locality-hours

Lower is better

PPO **20.2**

Fixed Rotation 102.1

0 — 168 locality-hours per episode

## Jain fairness

Higher is better

PPO **96.91%**

Fixed Rotation 87.76%

0% — 100%

## Utilization

Higher is better

PPO **97.61%**

Fixed Rotation 71.21%

0% — 100%

## The paired difference

Paired 95% intervals · pp = percentage points

| Metric | PPO | Fixed Rotation | PPO – Fixed Rotation |
|---|---|---|---|
| Critical-outage locality-hours | 20.2 locality-hours | 102.1 locality-hours | -81.9 [-82.5, -81.2] locality-hours |
| Jain fairness | 96.91% | 87.76% | +9.15 [9.11, 9.19] pp |
| Utilization | 97.61% | 71.21% | +26.39 [26.34, 26.45] pp |

Differences use unrounded values.

Power balance: maximum power-balance violation rate 0.00% across both controllers.

Comparison scope. Fixed Rotation is supply-blind, uses only Off and Full Service, and cannot select Critical Only.

Synthetic scenarios · Decision support only.
Frozen results stay separate from the live scenario.

Close

# Evidence details

Close

Frozen documented-seed-disjoint holdout. Live demonstrations cannot replace these results.

Evidence artifact

gridguard-robustness-supplement-v2

Release

gridguard-khartoum-v1.2.0

SHA-256

6b56d149331ee191302cd9e6f22365988db0ae301f9441190b61214acf832e48

**Hash verified · Frozen**

320 documented-seed-disjoint independent base draws × 3 shortage levels = 960 matched scenario conditions · 1,920 controller evaluations

## Interpreting the evidence

Base paired 95% confidence intervals appear in the Results table. Sensitivity: pointwise 95%; no multiplicity adjustment or result gate. There is no favorable/adverse release gate.

Seeds are disjoint only from every documented training, validation, and model-selection seed in data/model/training-provenance.yaml. The holdout was not preregistered or independently custodied and cannot rule out undocumented activity.

Comparator boundary: Fixed Rotation is supply-blind, limited to Off and Full Service, and cannot select Critical Only. Synthetic decision support only, not field evidence. No behavior-cloning-only ablation was run, so this comparison cannot isolate the incremental effect of PPO fine-tuning.

Sensitivity intervals are descriptive, pointwise 95%, and unadjusted for multiple comparisons; they do not support family-wide significance claims.

The power-balance result is a simulator conservation invariant, not evidence of real-grid operational safety. The accepted reproduction is historical parity evidence; its three seeds participated in model selection.

## Study limitations

Synthetic demand traces do not establish real-world forecast accuracy.

Sensitivity covers only one-factor-at-a-time ±25% demand-allocation perturbations.

Sensitivity intervals are descriptive, pointwise 95%, and unadjusted for multiple comparisons; they do not support family-wide significance claims.

Seeds are disjoint only from every documented training, validation, and model-selection seed in data/model/training-provenance.yaml. The holdout was not preregistered or independently custodied and cannot rule out undocumented activity.

No behavior-cloning-only ablation was run, so this comparison cannot isolate the incremental effect of PPO fine-tuning.

# Extended study

960 matched scenario conditions from 320 independent base draws; 2,880 controller evaluations.

Mild, Moderate and Severe shortage. Evaluation seeds: 91001, 91002, 91003, 91004, 91005, 91006, 91007, 91008, 91009, 91010. Batch size: 32.

Controllers | Paired differences | Forecasting

| Metric | Fixed Rotation | PPO | Optimization |
|---|---|---|---|
| Critical interruption (locality-hours) | 102.07 | 19.86 | 6.75 |
| Cumulative-service Jain fairness | 87.76% | 96.89% | 94.89% |
| Supply utilization | 71.24% | 97.62% | 98.24% |
| Critical energy served | 42.63% | 99.70% | 99.87% |
| Zero-service blackout (locality-hours) | 96.00 | 0.00 | 0.00 |
| Blackout-duration disparity (hours) | 1.71 | 0.00 | 0.00 |
| Critical-duration disparity (hours) | 1.90 | 0.00 | 0.00 |
| Longest blackout per episode (hours) | 6.00 | 0.00 | 0.00 |

Optimization exhaustively maximizes the canonical one-hour reward over 2,187 service-tier actions. It uses current and past state; it is not a 24-hour optimum.

Blackout means zero service when demand is positive. Critical interruption means less than 99% of critical demand served. Duration disparity is the mean gap across the 21 locality pairs within each episode; equal durations can still mean poor service. Global proportional clipping can give every locality the same critical interruption hours.

Synthetic scenarios; decision support only.
Evidence: gridguard-proposal-supplement-v1

Close

# Extended study

960 matched scenario conditions from 320 independent base draws; 2,880 controller evaluations.

Mild, Moderate and Severe shortage. Evaluation seeds: 91001, 91002, 91003, 91004, 91005, 91006, 91007, 91008, 91009, 91010. Batch size: 32.

Controllers | **Paired differences** | Forecasting

Compare: PPO - Fixed Rotation

| Metric | Mean difference | Paired 95% interval |
|---|---|---|
| Critical interruption (locality-hours) | -82.22 locality-h | [-82.91, -81.53] locality-h |
| Cumulative-service Jain fairness | +9.13 pp | [+9.10, +9.17] pp |
| Supply utilization | +26.38 pp | [+26.33, +26.43] pp |
| Blackout-duration disparity (hours) | -1.71 h | [-1.71, -1.71] h |
| Critical-duration disparity (hours) | -1.90 h | [-1.90, -1.90] h |

Difference = first controller minus second. Lower interruption durations and duration gaps are better; higher Jain fairness and utilization are better. pp means percentage points. Intervals are descriptive and pointwise.

Synthetic scenarios; decision support only.
Evidence: gridguard-proposal-supplement-v1

Close

# Extended study

960 matched scenario conditions from 320 independent base draws; 2,880 controller evaluations.

Mild, Moderate and Severe shortage. Evaluation seeds: 91001, 91002, 91003, 91004, 91005, 91006, 91007, 91008, 91009, 91010. Batch size: 32.

Controllers | **Paired differences** | Forecasting

Compare: Optimization - Fixed Rotation

| Metric | Mean difference | Paired 95% interval |
|---|---|---|
| Critical interruption (locality-hours) | -95.32 locality-h | [-95.72, -94.91] locality-h |
| Cumulative-service Jain fairness | +7.13 pp | [+7.05, +7.21] pp |
| Supply utilization | +27.00 pp | [+26.95, +27.05] pp |
| Blackout-duration disparity (hours) | -1.71 h | [-1.71, -1.71] h |
| Critical-duration disparity (hours) | -1.90 h | [-1.90, -1.90] h |

Difference = first controller minus second. Lower interruption durations and duration gaps are better; higher Jain fairness and utilization are better. pp means percentage points. Intervals are descriptive and pointwise.

Synthetic scenarios; decision support only.
Evidence: gridguard-proposal-supplement-v1

Close

# Extended study

960 matched scenario conditions from 320 independent base draws; 2,880 controller evaluations.

Mild, Moderate and Severe shortage. Evaluation seeds: 91001, 91002, 91003, 91004, 91005, 91006, 91007, 91008, 91009, 91010. Batch size: 32.

Controllers | **Paired differences** | Forecasting

Compare: PPO - Optimization

| Metric | Mean difference | Paired 95% interval |
|---|---|---|
| Critical interruption (locality-hours) | +13.10 locality-h | [+12.32, +13.86] locality-h |
| Cumulative-service Jain fairness | +2.00 pp | [+1.93, +2.08] pp |
| Supply utilization | -0.62 pp | [-0.65, -0.59] pp |
| Blackout-duration disparity (hours) | +0.00 h | [+0.00, +0.00] h |
| Critical-duration disparity (hours) | +0.00 h | [+0.00, +0.00] h |

Difference = first controller minus second. Lower interruption durations and duration gaps are better; higher Jain fairness and utilization are better. pp means percentage points. Intervals are descriptive and pointwise.

Synthetic scenarios; decision support only.
Evidence: gridguard-proposal-supplement-v1

Close

# Extended study

960 matched scenario conditions from 320 independent base draws; 2,880 controller evaluations.

Mild, Moderate and Severe shortage. Evaluation seeds: 91001, 91002, 91003, 91004, 91005, 91006, 91007, 91008, 91009, 91010. Batch size: 32.

Controllers | Paired differences | Forecasting

## One-hour-ahead demand prediction

Learned forecast fitted on 768 training traces from seeds 42, 43, 44. 320 evaluation traces use the separate study seeds above, targeting hours 3–23.

| Forecast method | Locality-sector MAE (MW) | Locality-sector RMSE (MW) | System total MAE (MW) | System total RMSE (MW) |
|---|---|---|---|---|
| Trailing mean (3 hours) | 2.3385 | 3.7436 | 42.7512 | 49.9759 |
| Persistence (1 hour) | 1.3465 | 2.1776 | 24.3809 | 28.5917 |
| Learned ridge AR(3) | 0.6573 | 1.1882 | 9.3376 | 11.9242 |

MAE is mean absolute error; RMSE is root mean square error. Lower is better. Locality-sector errors and total-system errors use different aggregation scales. This forecast comparison does not retrain or alter the PPO policy.

Synthetic scenarios; decision support only.
Evidence: gridguard-proposal-supplement-v1

Close